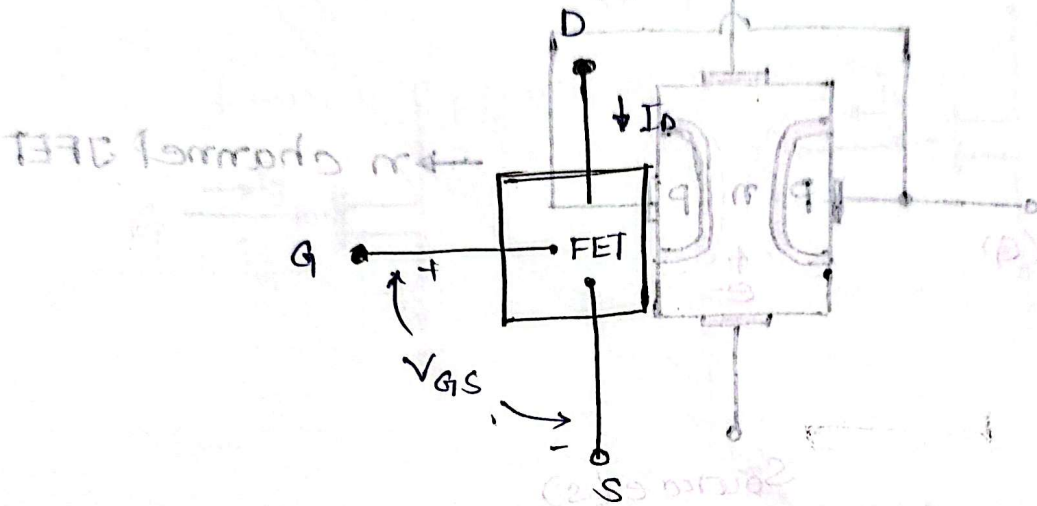


Field-Effect Transistor

→ BJ ← 3 terminal ← FET



* FET → voltage controlled device

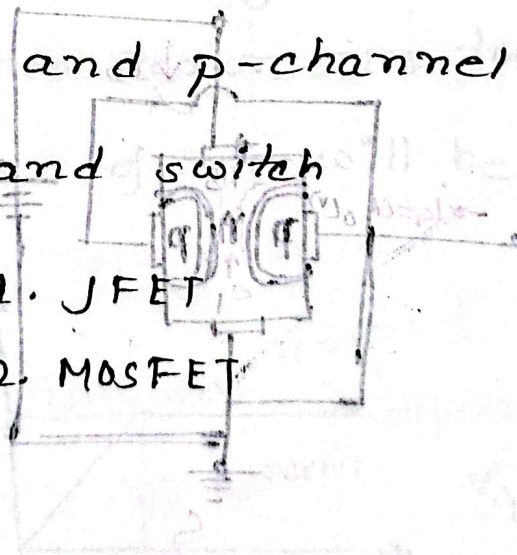
$$I_D = f(V_{GS})$$

* FET → unipolar

* n channel and p-channel.

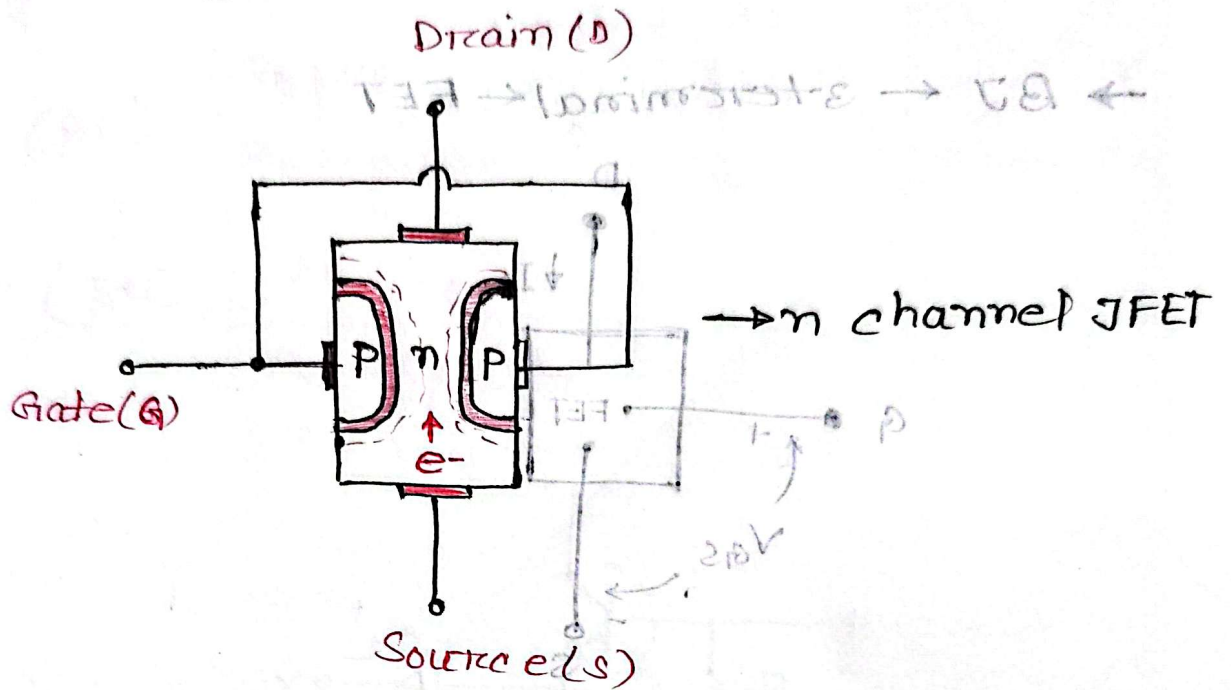
* FET → amp and switch

* two types : 1. JFET
2. MOSFET



$$V_{DS} = 20V$$

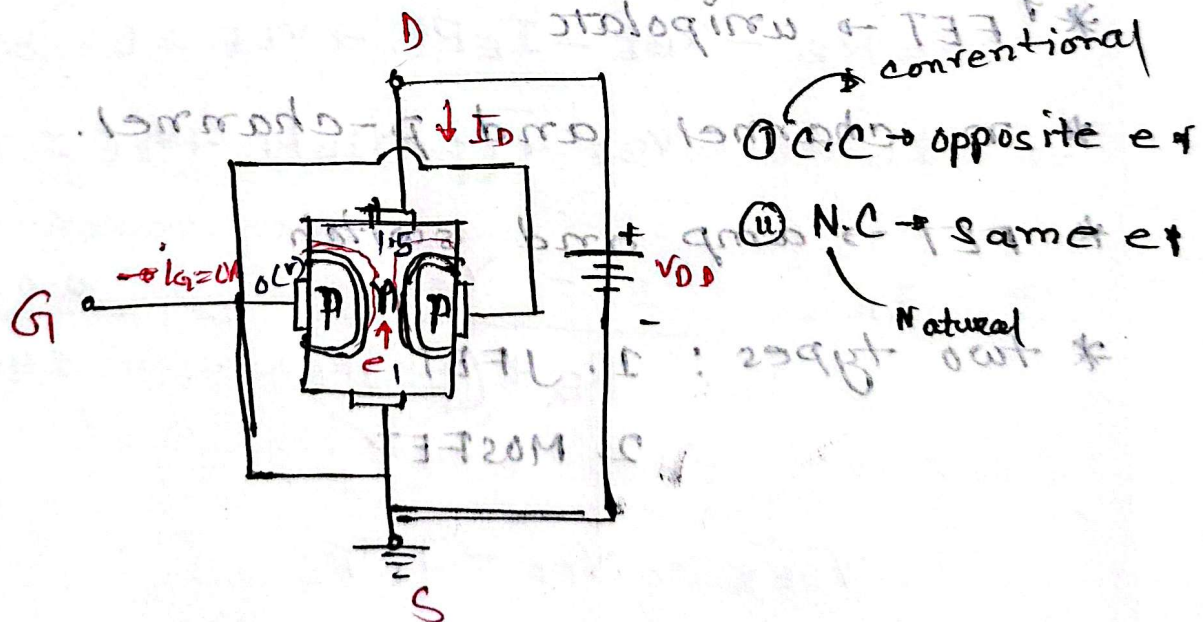
JFET



Case 1: $V_{GS} = 0V$
 Case 2: $V_{GS} < 0V$

$V_{DS} > 0V$

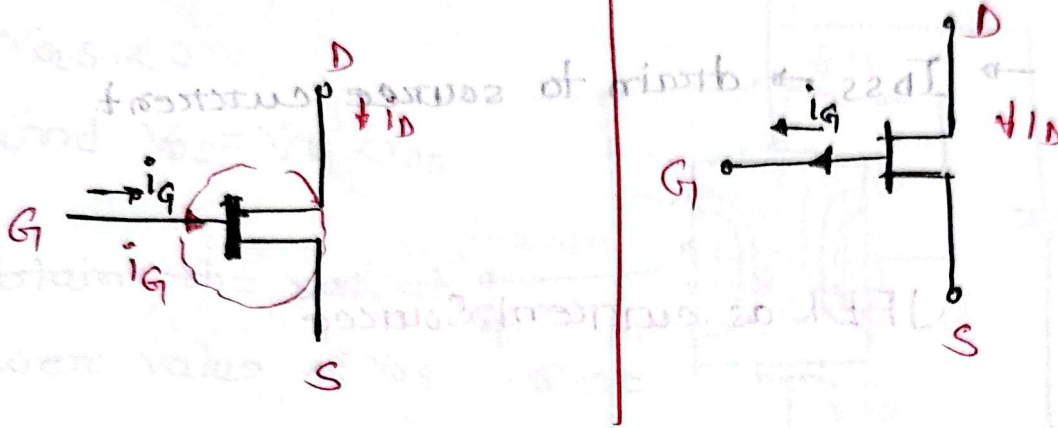
Case-1



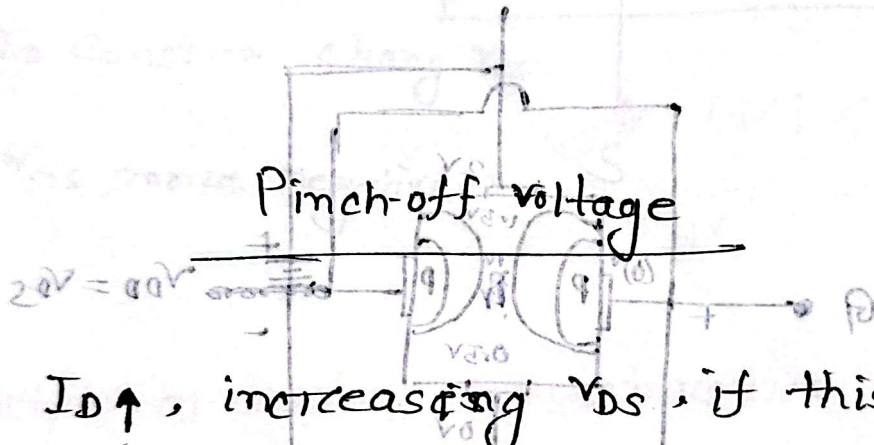
$$V_{DS} = V_{DD}$$

symbol:

n channel $V_D = 20V$ p channel



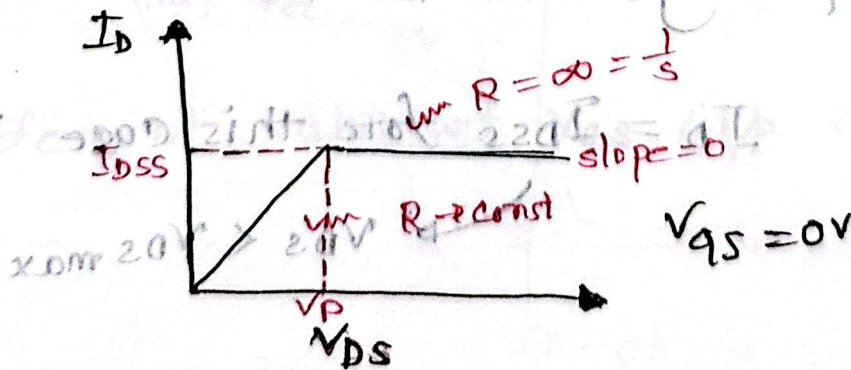
Pinch-off voltage



$V_{DS} \uparrow$ $I_D \uparrow$, increasing V_{DS} , if this happens

that two depletion region are touching each other

then the value of V_{DS} will be pinch off voltage



I_{DSS} $\rightarrow$ maximum drain current

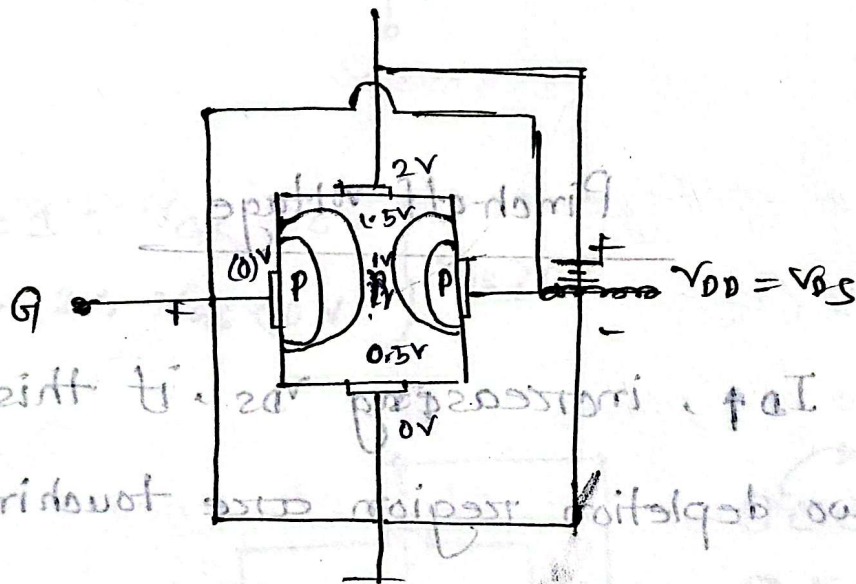
↓

$$N_G S = 0$$
$$V_{GS} = 0 \rightarrow \boxed{V_{GS} = 0V}, \boxed{V_{DS} > |V_P|}$$

also $\rightarrow I_{DSS} \rightarrow$ drain to source current

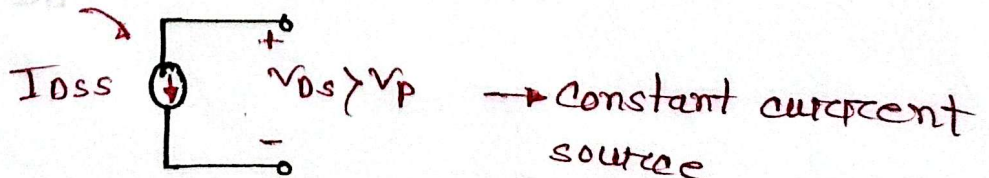


JFET as current Source



so $p \rightarrow n$ junction reverse biased

$I_D = I_{DSS}$ for this case $v_{DS} > v_p$

$$v_0 = 2\rho v$$
$$V_{DS} < V_{DS \max}$$


working of JFET (negative Voltage at the Gate)

Case 2:

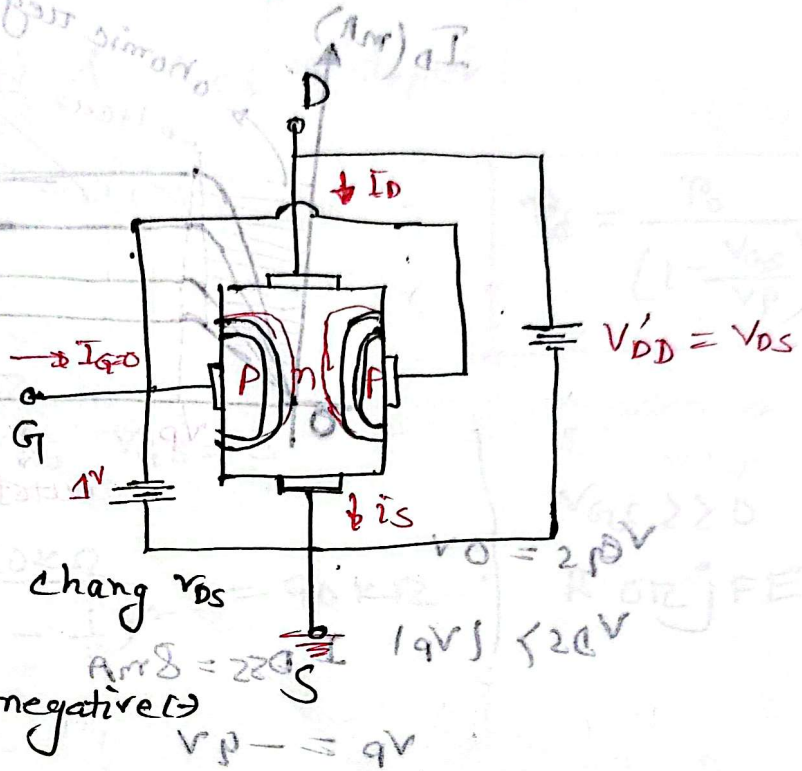
$$V_{GS} < 0$$

$$\text{and } V_{DS} = V_{DD}' < V_{DD}$$

obtain the sat. at
lower value of V_{DS}

sat $\rightarrow I_D \text{ Const} \rightarrow \text{chang } V_{DS}$

V_{GS} more negative



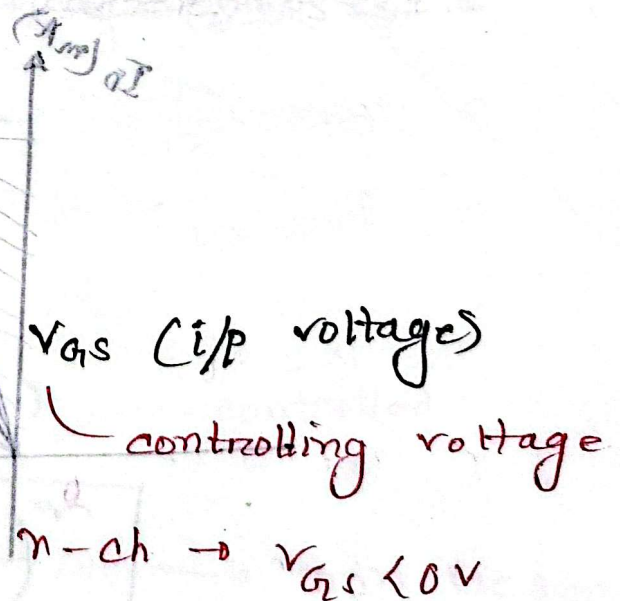
Output of Drain characteristics of JFET

JFET

I_D V_{GS} V_{DS}
outp. cu. outp. vot.

for different value of V_{GS} (i/p voltage)

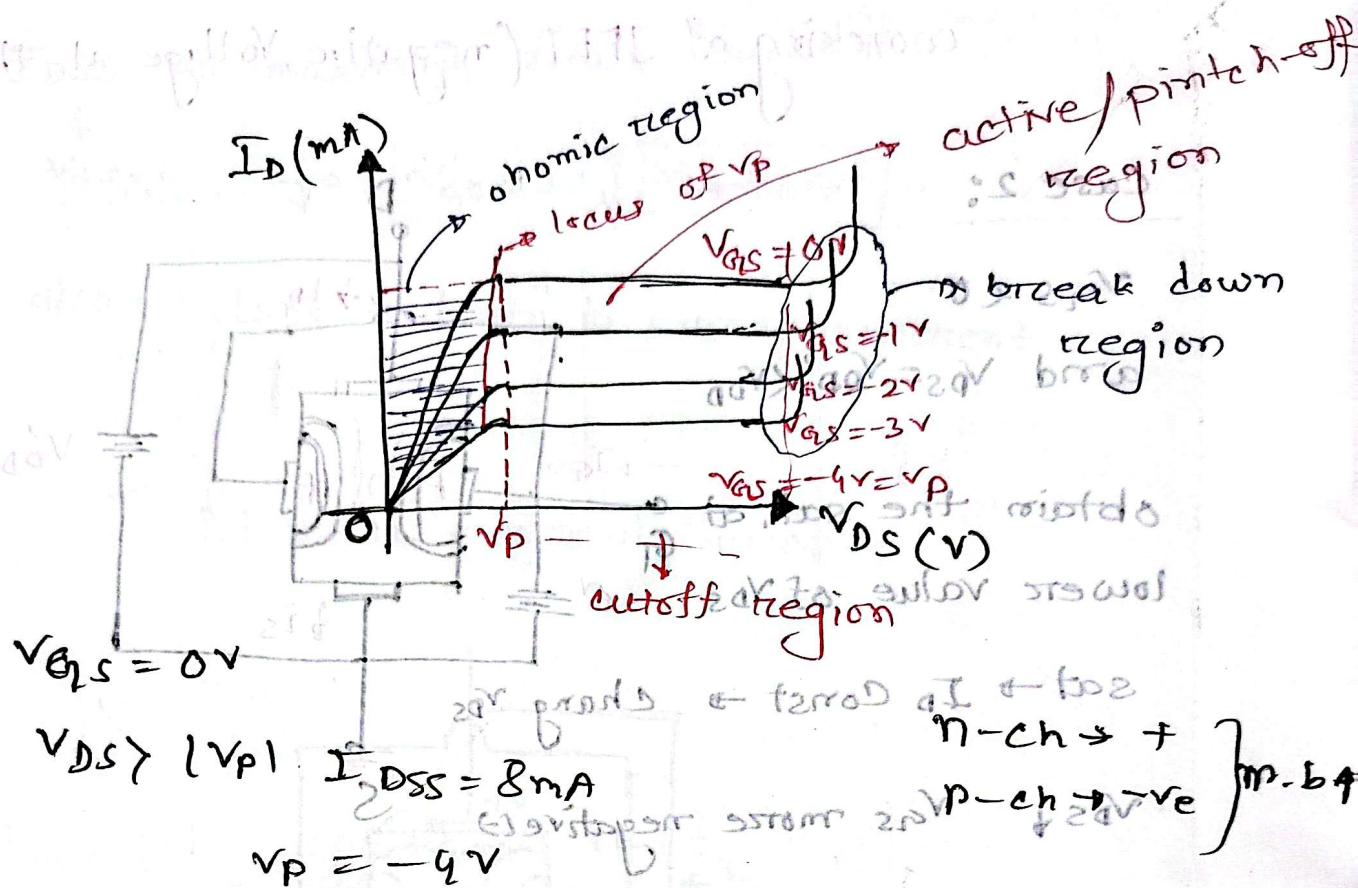
(v) 20V



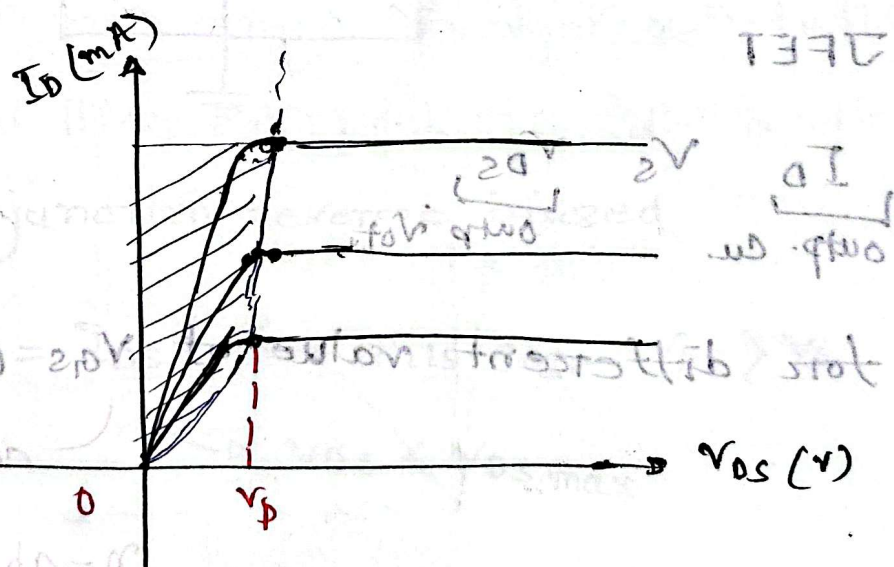
n-ch $\rightarrow V_{GS} < 0V$

$V_{GS} > 0V$

JFET - variable resistor



JFET as a Voltage-Controlled Resistor



ohmic region $\rightarrow$ volt. controlled. Resi. $R_{DS(on)}$

JFET $\rightarrow$ variable resistor

$R \uparrow$ or $\downarrow$

slope $\Rightarrow f(V_{GS})$ & $V_{GS} \downarrow \Rightarrow \text{slope} \downarrow$

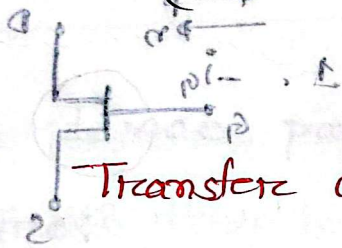
$$R = 1/\text{slope} \Rightarrow R = f(V_{GS})$$

$$r_d = \frac{r_o}{\left(1 - \frac{V_{GS}}{V_P}\right)^2}$$

Ex: $r_o = 10k\Omega$
 $V_P = -6V$, $V_{GS} = -3$

$$r_d = \frac{10k\Omega}{\left(1 - \frac{-3}{-6}\right)^2} = 40k\Omega$$

$V_{GS} \gg 0$
 R or JFET $\uparrow$



Transfer characteristics of JFET

what is transfer characteristics
 plot I_D vs V_{GS} (with $V_{DS} \rightarrow \text{const}$)

JFET: $I_D = f(V_{GS})$

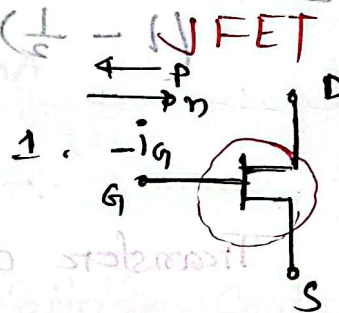
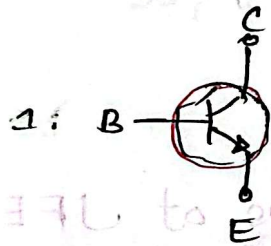
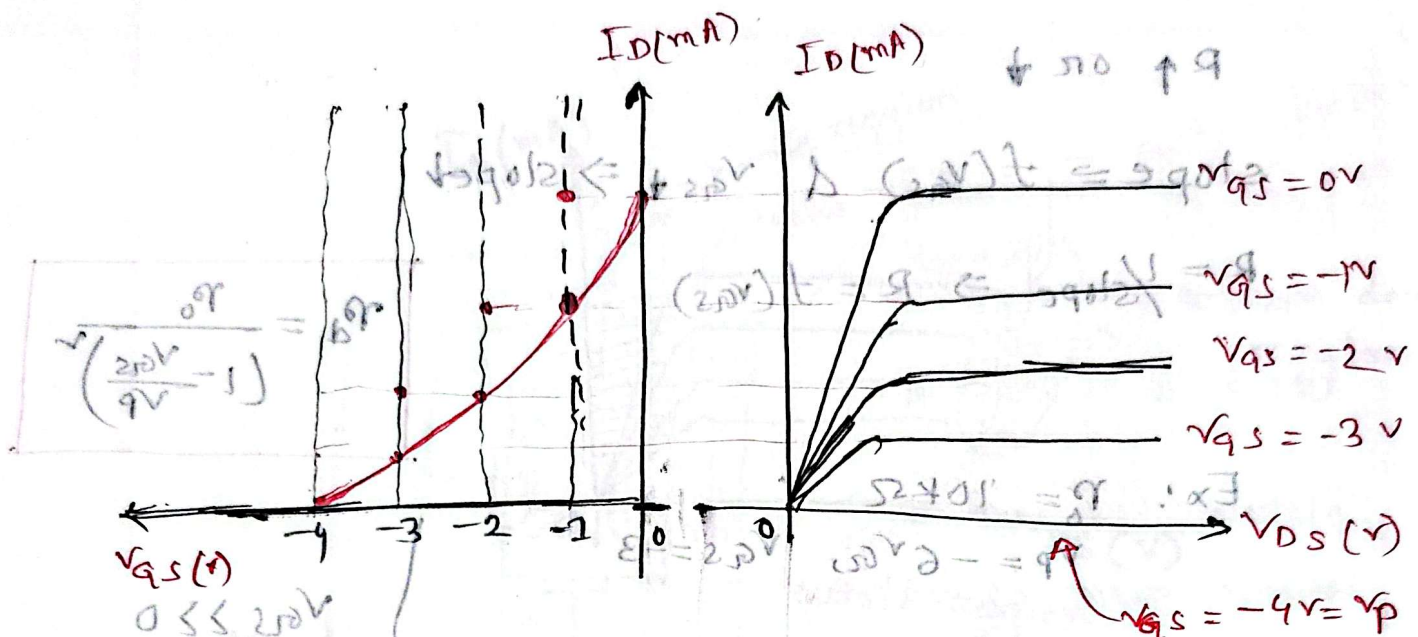
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

controlled

non linear

const

$V_{GS} \downarrow = I_D \uparrow$ (expon.)



2. $I_C = f(I_B) = \beta I_B$

2. $I_D = f(V_{GS}) = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$

3. Current controlled

3. Voltage controlled

4. Bipolar

4. Unipolar $\rightarrow e^-$ or holes

5. $I_C \approx I_E$

5. $I_D = I_S$

6. $V_{BE} \approx 0.7$

6. $I_G = 0A$

7. Good fanout

7. Bad fanout

8. linear amplification

8. non-linear amplification

9. power consumption

9. power cons

10. $Z_i^{BJT} < Z_i^{JFET}$

$Z_i \rightarrow$ input impedance

- | | |
|------------------------------|---------------------------------|
| 11. Bad thermal stability | 11. Good thermal stability |
| 12. Gain $\uparrow$ | 12. Gain $\downarrow$ = 20V |
| 13. sensitivity $\downarrow$ | 13. Sensitivity $\uparrow$ 10T |
| 14. noise level $\uparrow$ | 14. noise level $\downarrow$ |
| 15. Size $\uparrow$ | 15. Size $\downarrow$ (smaller) |

JFET problems

Ans The device parameters for n-channel JFET are:

maximum drain current (I_{DSS}) = 10 mA

pinch off voltage $\rightarrow V_P = -4V$

calculate drain current: $V_S = 20V$

a) $V_{GS} = 0V$, b) $V_{GS} = -1V$, c) $V_{GS} = -4V$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

a) $V_{GS} = 0$ & $V_{DS} > |V_P|$

$$I_D = I_{DSS} = 10 \text{ mA}$$

Case b: $V_{GS} = 1V$

$$V_{GS} = 1V$$

$$I_D = 10mA \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$I_D = 5.625mA$$

Case c: $V_{GS} = -4V$

$$V_{GS} = -4V = V_P$$

$$I_D = 0A$$

problem 2:

A JFET produces gate current of 2nA when it is reverse biased with 8V. Determine the resistance between gate and source.

$$I_G = 2nA$$

$$V_{GS} = 8V$$

$$V_{GS} = I_G \times R_{GS}$$

$$\Rightarrow R_{GS} = V_{GS} / I_G = 8V / 2 \times 10^{-9} = 4 \times 10^9 \Omega$$

$$|V_{GS}| < |V_P| \quad \& \quad V_P = 20V$$

$$I_D = 20mA = 0.1A$$

part - 2 (problem)

problem 1: A reverse gate voltage of JFET

when changes from 4.4V to 4.2V , the drain current changes from 2.2mA to 2.6mA . Find out the value of transconductance.

Soln:

$$g_m = \frac{\text{outp } i}{\text{inp. } V} \quad \left| \text{for const. outp. } V \right.$$

$$g_m = \frac{\text{change in output curr.}}{\text{change in input. voltage}} \quad \left| \text{const. outp. } V \right.$$

$$g_m = \frac{4.4 - 4.2 \text{ (} \Delta V_{GS} \text{)} (2.6 - 2.2) \text{mA}}{(2.6 - 2.2) \text{mA} (4.4 - 4.2) \text{V}}$$
$$= 2 \text{ mA/V or } 2 \text{ m mho}$$

problem 2: (multiple choice)

The pinch off voltage for an n-channel JFET is 4 volts, then pinch off occurs for V when V

is -4 volts if V_{GS} is 1 volt

two biasing of JFET $V_P = 4V$ & $V_{GS} = 1V$

a) 3 volts

b) 5 "

c) 4 "

d) 1 "

$$|V_{DS}| = |V_P| - |V_{GS}|$$

$$= 4 - 1 = 3V$$

$$\frac{I_{DSS}}{V_{GS}} = \mu B$$

$$\frac{I_{DSS} - I_D}{V_{GS}} = \mu B$$

$$I_D = \frac{I_{DSS}(V_{GS} - V_P)^2}{V_P^2}$$

$$V_{GS} = 1V \quad I_D = 5mA$$

$$V_{GS} = I_D R_{GS}$$

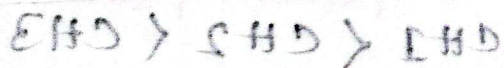
$$R_{GS} = \frac{V_{GS}}{I_D} = \frac{1V}{5mA} = 200\Omega$$

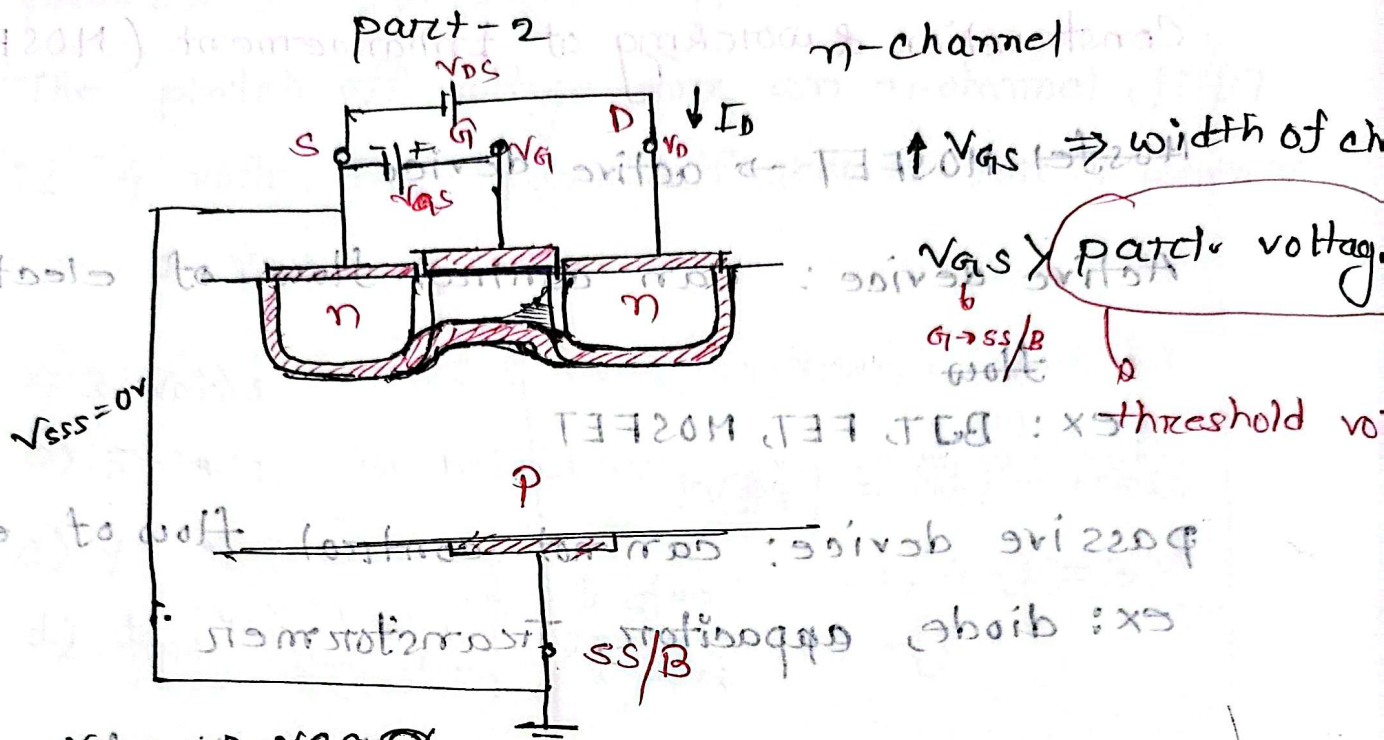
~~Mosfet~~ MOSFET $\rightarrow$ active device

Active device : can control flow of electron flow

passive device: can not control flow of electron
ex: diode, capacitor, transformer

Enhancement type $\rightarrow$ n-ch
p-ch

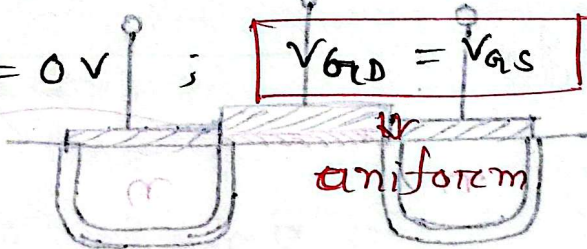




$$V_{GD} = V_{GS} + V_{DS} = V_D$$

$$\Rightarrow \boxed{V_{GD} - V_D = 0 \text{ } \& \text{ } V_{GS} - V_{DS} = V_{GD}}$$

Case 1: $V_{DS} = 0V$; $\boxed{V_{GD} = V_{GS}}$



$$V_T = 1V$$

$$V_{GS} = 2V = V_{GD}$$

$$\boxed{\text{Excess vol} = V_{GS} - V_T}$$

$$V_{GS} = 3V = V_{GD}$$

$$V_{GS} = 4V = V_{GD}$$

$$\rightarrow \text{ch}_1 \rightarrow 1V$$

$$\rightarrow \text{ch}_2 \rightarrow 2V$$

$$\rightarrow \text{ch}_3 \rightarrow 3V$$

$$; \boxed{\text{ch}_1 < \text{ch}_2 < \text{ch}_3}$$

Case 2: $V_{DS} > 0V$

$$V_{GD} \neq V_{GS}$$

$$D_{rain} \rightarrow \oplus V_C$$

Case-3:

$$V_{DS} = V_{GS} - V_T$$

$$V_{GD} = V_T$$

Drain & Transfer characteristics

output
ch.

$$o/p \ i \rightarrow I_D \quad V_{DS} \leq V_{GS} - V_T$$

$$o/p \ v \rightarrow V_{DS} \quad V_{DS} > V_{GS} - V_T$$

Drain char $\rightarrow I_D$ vs V_{DS} For various levels

$$V_T \& K \quad \frac{(m) \mu I}{(V - (m) 2V)}$$

$$C.1: V_{GS1} > V_T$$

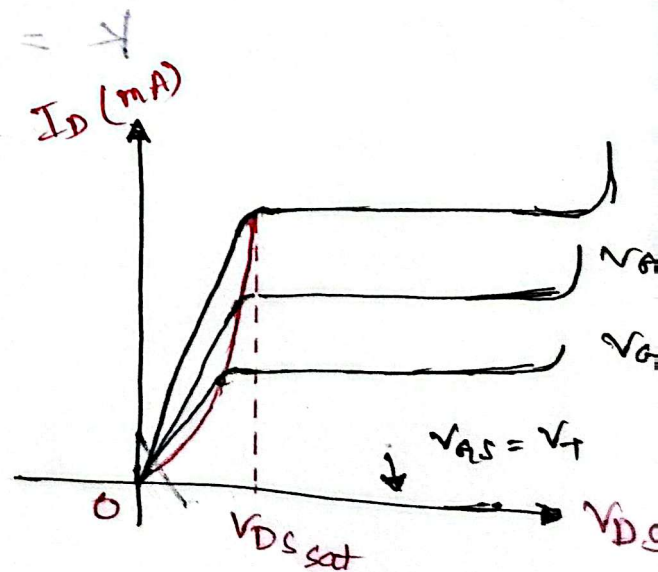
$$V_{CEFF} = V_{GS1} - V_T$$

$$V_{DS} = 0 \Rightarrow V_D = V_S$$

$$V_{DS} \uparrow \Rightarrow I_D \uparrow$$

$$V_{DS} = V_{DS \text{ sat}} \Rightarrow$$

$$I_D = \text{const. (pinch-off)}$$



in case of pinch-off:

$$V_{DS} = V_{GS} - V_T$$

$$V_{GD} = V_T$$

Case - 1):

$$V_{GS2} < V_{GS1}$$

$$P_2 < P_1$$

$$R_2 > R_1$$

slope 2 < slope

$$V_{DS sat} = V_{GS2} - V_T$$

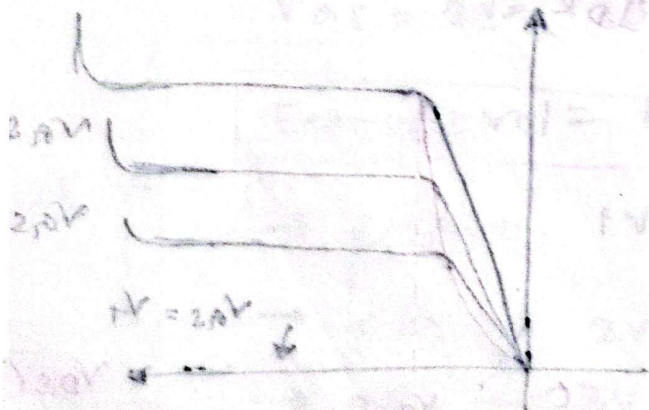
Sat - Reg:

$$V_{DS} \geq V_{GS} - V_T$$

$$I_D = k (V_{GS} - V_T)^2$$

constant

$$k = \frac{I_D (on)}{(V_{GS(on)} - V_T)^2}$$



MOSFET - Problem 11

1. Does the $\frac{(I_D)_E}{(I_D)_D}$ of enhancement type MOSFET increases at the same rate as a depletion-type MOSFET for the conduction region?

Enha:

$$I_D = k(V_{GS} - V_T)^2$$

Const: k and V_T

$$\frac{dI_D}{dV_{GS}} = 2kV_{GS} - 2kV_T$$

$$\frac{dI_D}{dV_{GS}} = k \cdot \frac{d(V_{GS} - V_T)^2}{dV_{GS}}$$

$$= 2kt \cdot \frac{d(V_{GS} - V_T)}{dV_{GS}}$$

$$= 2k(V_{GS} - V_T) \cdot \frac{d(V_{GS} - V_T)}{dV_{GS}}$$

$$\frac{dI_D}{dV_{GS}} = 2k(V_{GS} - V_T)$$

$$= C(V_{GS} - C)$$

Dep:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Const: I_{DSS} and V_P

$$\frac{dI_D}{dV_{GS}} = \frac{2I_{DSS}}{V_P} + \frac{2I_{DSS}V_{GS}}{V_P^2}$$

$$= I_{DSS} \frac{d}{dV_{GS}} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= I_{DSS} \frac{d}{dV_{GS}} \cdot S^2$$

$$= 2I_{DSS} \cdot S \cdot \frac{dS}{dV_{GS}}$$

$$= 2I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)$$

$$\frac{d}{dV_{GS}} \left(1 - \frac{V_{GS}}{V_P}\right)$$

$$\frac{dI_D}{dV_{GS}} = \frac{2I_{DSS}}{V_P} (V_{GS} - V_P)$$

$$= C(V_{GS} - C)$$

MOSFET problem 21

Enhance:

1. Given $k = 0.4 \times 10^{-3} \text{ A/V}^2$ and $I_{D(\text{on})} = 3.5 \text{ mA}$ with $V_{GS(\text{on})} = 4 \text{ V}$, determine V_T .

$$I_D = k (V_{GS} - V_T)^2$$

$$(2 \text{ V} - 1) 220 \text{ I} = 0 \text{ I}$$

$$\Rightarrow 3.5 \times 10^{-3} = 0.4 \times 10^{-3} (4 - V_T)^2$$

$$\Rightarrow V_T = 1.092 \text{ V}$$

Triode

$$I_D = 2k [(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2}]$$

$$2 \cdot \frac{b}{2 \mu \text{ V b}} 220 \text{ I} =$$

$$\frac{b}{1 \text{ b}} 2 \cdot 220 \text{ I} =$$

$$\frac{2 \text{ V}}{1 \text{ V}} - 1) 220 \text{ I} =$$

$$- 1) \frac{b}{2 \mu \text{ V b}}$$

Deple:

Given $I_D = 4 \text{ mA}$ at $V_{GS} = -2 \text{ V}$, determine the saturation current if $V_P = -4 \text{ V}$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$4 \times 10^{-3} = I_{DSS} \left(1 - \frac{-2}{-4} \right)^2$$

$$\Rightarrow I_{DSS} = 8 \times 10^{-3} \text{ A}$$

$$\frac{I_D}{2 \mu \text{ V b}} = \frac{b}{2 \mu \text{ V b}} \cdot \frac{V_{GS} - V_T}{2} = \frac{0 \text{ I b}}{2 \mu \text{ V b}}$$

$$\frac{I_D}{2 \mu \text{ V b}} \cdot \frac{b}{2 \mu \text{ V b}} =$$

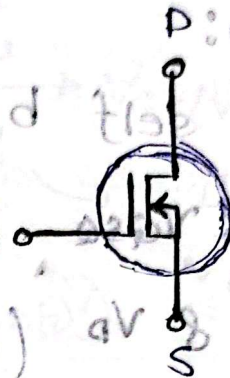
$$\frac{b}{2 \mu \text{ V b}} \cdot \frac{b}{2 \mu \text{ V b}} =$$

$$(V - 2 \text{ V}) \cdot \frac{b}{2 \mu \text{ V b}} = \frac{0 \text{ I b}}{2 \mu \text{ V b}}$$

MOSFET biasing

Simila JFET

- (ମିଶ୍ର ଓ)



→ n-channel

→ D-MOSFET

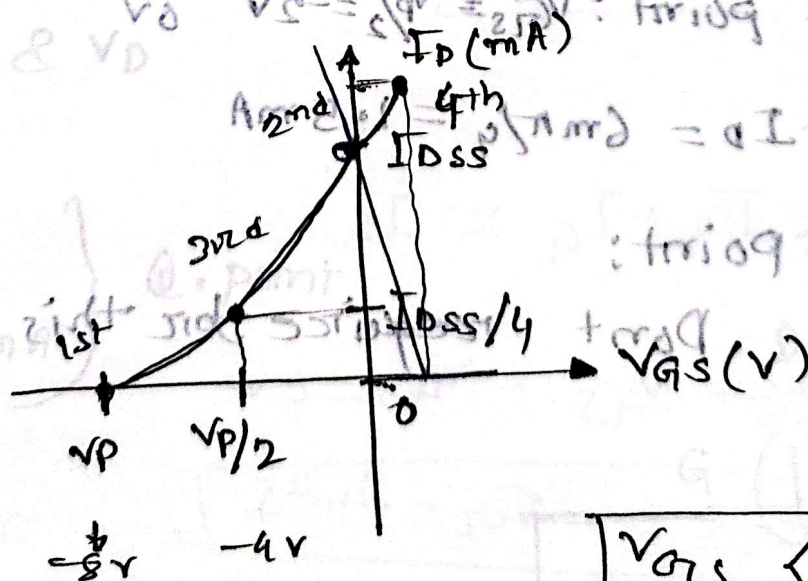
$$\rightarrow I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$V_{GS} < 0 \text{ V}$$
$$V_{GS} = 0V$$

Loss $\neq$ Max

$$V_4 S \{ 0 V | X$$
$$V_{GS} < 0V$$
$$V_{GS} = 0V$$

✓GIS yov

$$I_{DSS} (\chi_{\max})$$

$$V_{GS} < +4V$$

Self bias Configuration of Depletion Type MOSFET

problem:

For the self bias-configuration, determine

a) I_{DQ} & V_{GSQ} (Q. point)

b) V_{DS} & V_D (op. v)

$$V_{GS} > 0V$$

1st pt: $I_D = 0A$

$$V_{GS} = -9V$$

2nd pt: $V_{GS} = 0$

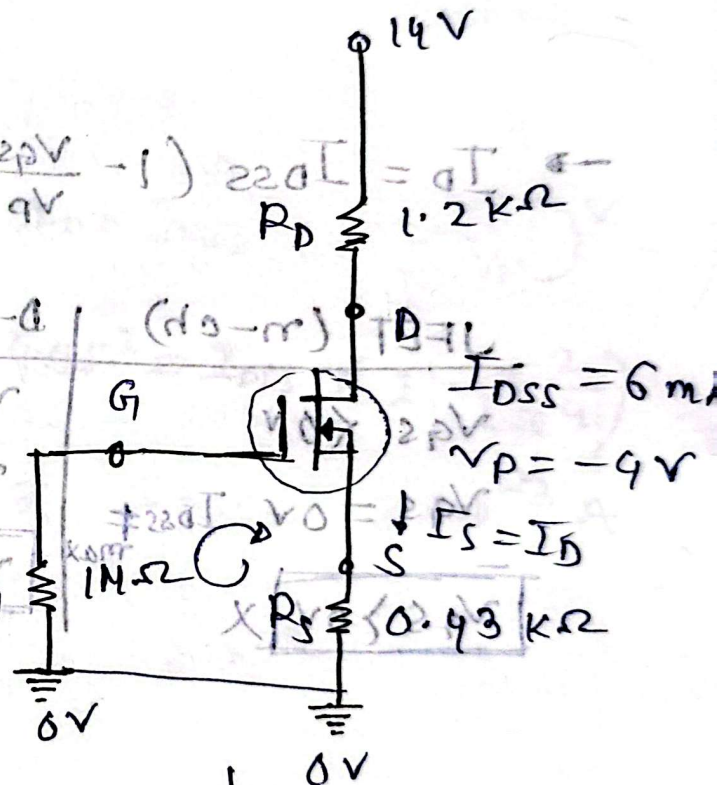
$$I_D = 6mA$$

3rd point: $V_{GS} = V_P/2 = -2V$

$$I_D = 6mA/9 = 1.5mA$$

4th point:

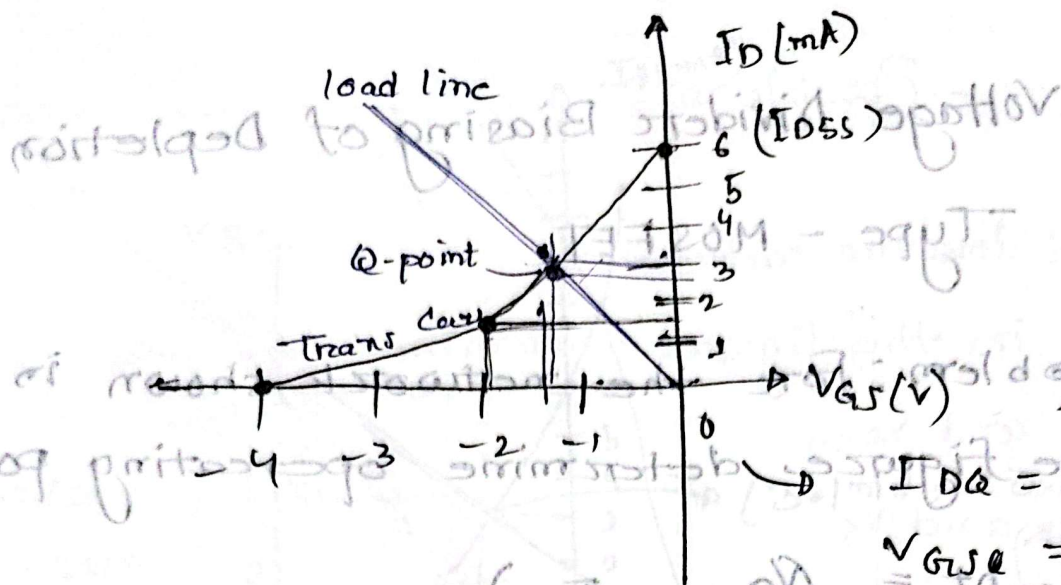
Don't require for this prob.



$$0 - V_{GS} - I_D R_S = 0V$$

$$\Rightarrow V_{GS} = -I_D R_S$$

$$V_P > 250V$$



$I_{DQ} = 2.8 \text{ mA}$
 $V_{GSQ} = 1.2 \text{ V}$

Forc load line

$I_D = \frac{I_{DSS}}{2} = 3 \text{ mA}$

$V_{GS} = -(3 \text{ mA}) \times (0.43 \text{ k}\Omega)$

$V_{GS} \approx -1.3 \text{ V}$

Depletion type MOSFET Biasing (Special-Case)

Problem: Forc the network determine a) Q point.

b) V_{DS} & V_D

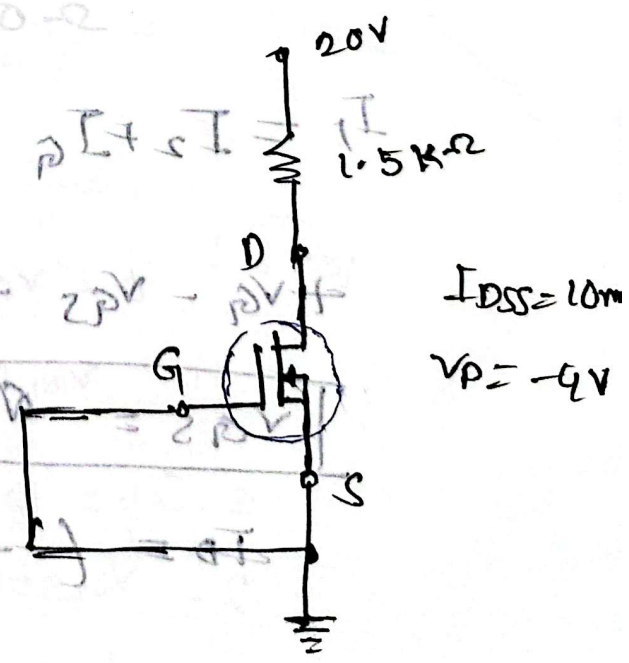
$V_{GSQ} = 0 \text{ V}$
 $I_{DQ} = 10 \text{ mA}$ } Q. point

Sol. b)

$20 - (1.5 \times 10) = V_{DS} = 5 \text{ V}$

$V_{DS} = 5 \text{ V}$

$V_{DS} = V_D - V_S \Rightarrow V_D = V_{DS} + V_S = 5 + 0 = 5 \text{ V (Ans)}$

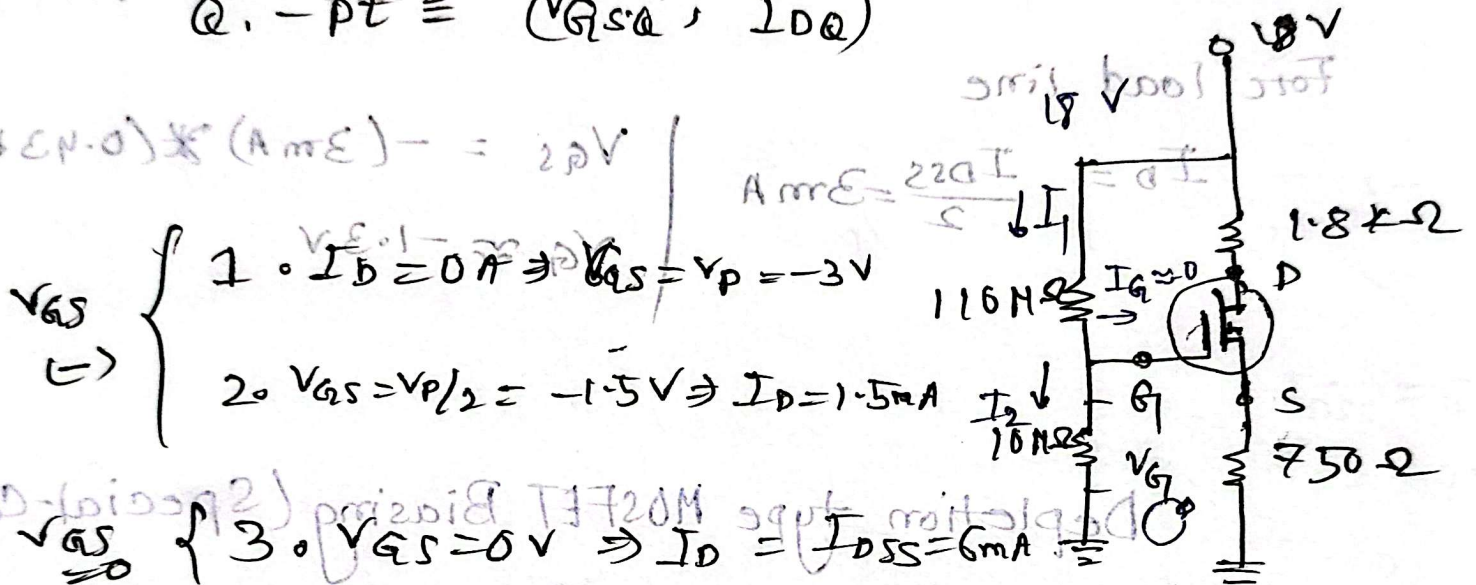


Voltage Divider Biasing of Depletion

Type - MOSFET

problem: For the network shown in the figure, determine operating point

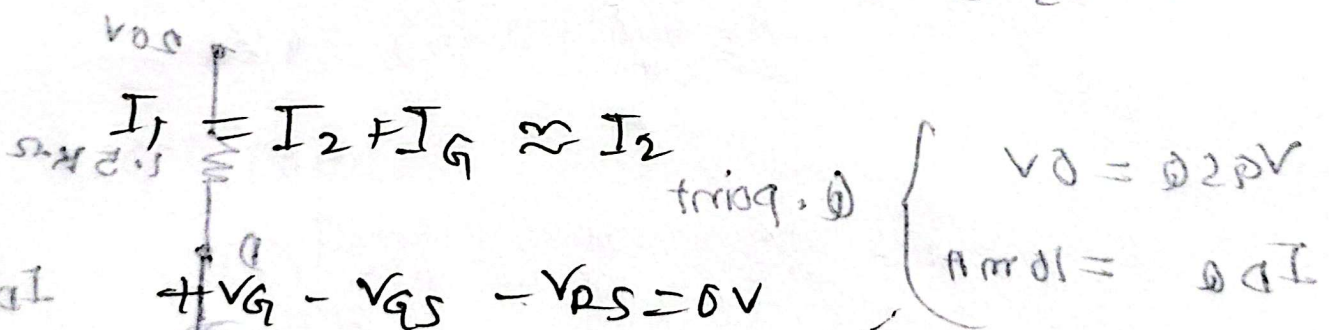
Q. - pt $\equiv (V_{GSQ}, I_{DQ})$



Handwritten notes for V_{GS} :

- 1. $I_D \geq 0A \Rightarrow V_{GS} = V_p = -3V$
- 2. $V_{GS} = V_p/2 = -1.5V \Rightarrow I_D = 1.5mA$
- 3. $V_{GS} = 0V \Rightarrow I_D = I_{DSS} = 6mA$

$V_{DD} = 10V, V_p = -3V$

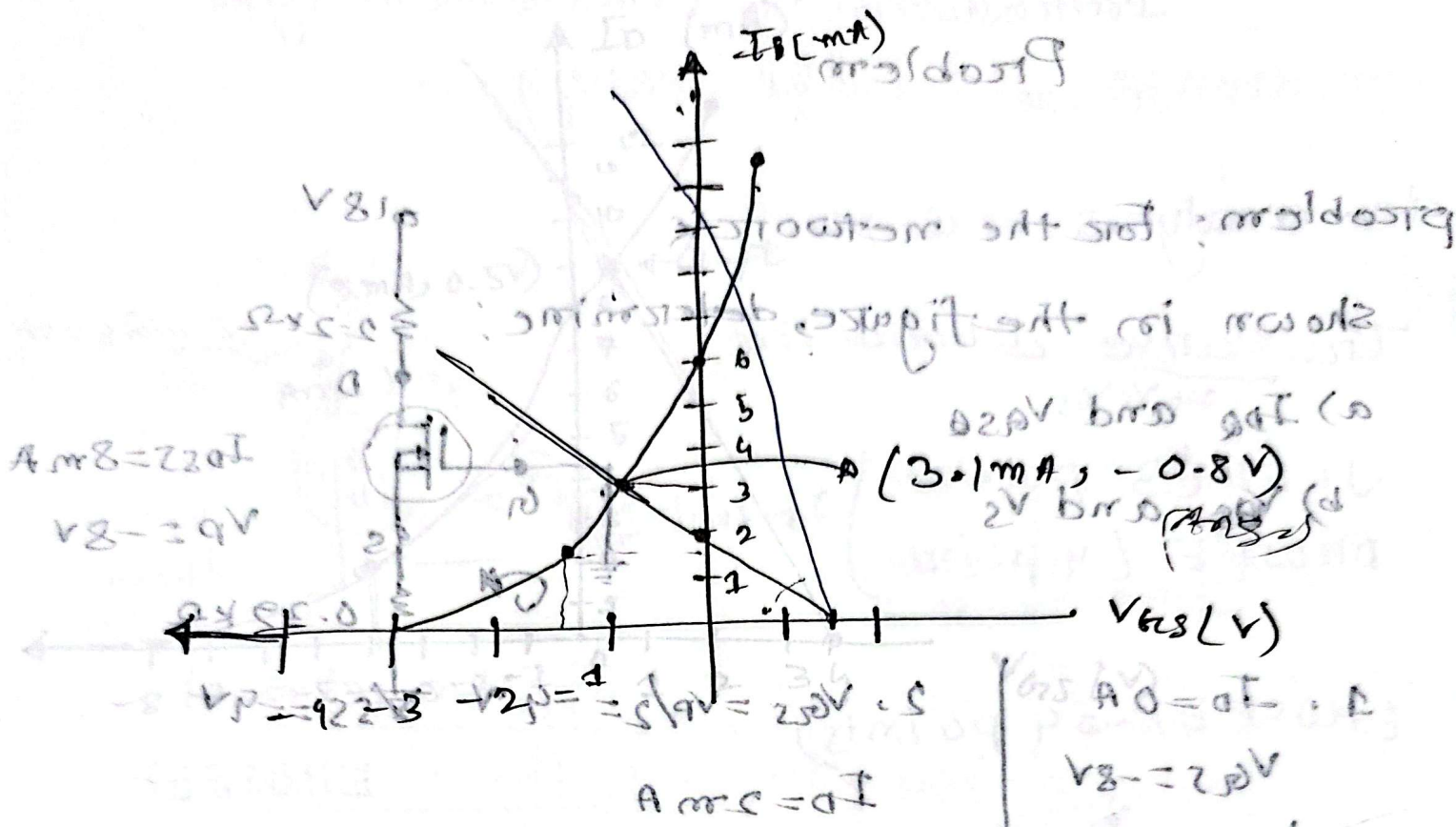


$$V_{GS} = V_G - I_D R_S$$

$$I_D = \left(-\frac{1}{R_S} \right) V_{GS} + \frac{V_G}{R_S}$$

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2} \approx 1.5V$$

$$\frac{V_G}{R_S} = \frac{1.5}{750\Omega} \approx 2mA$$



for load line; $I_D = 0A (-) = 2.2V$

$$V_{GS} = V_G = 1.5V$$

$$V_{GS} = 2.2V$$

$$V_{GS} = 2.2V$$

$$I_D = 1.5 \text{ mA}$$

$$V_{GS} = V_G - I_D R_S$$

$$V_{GS} = 2.2V$$

next part:

$$R_S = 150 \Omega$$

$$V_P = 0 = 2.2V - I_D R_S - 0$$

$$V_P = 2.2V, I_D = 0.1 \text{ mA}$$

$$I_D = 0.1 \text{ mA}, V_P = 2.2V$$

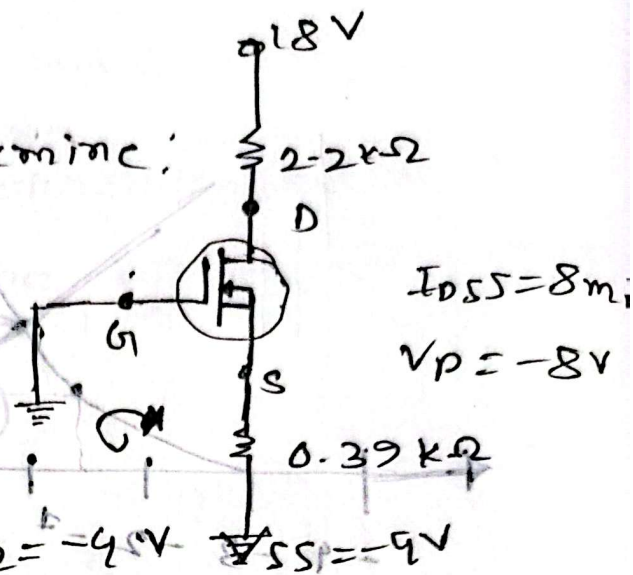
Problem

problem: For the network

shown in the figure, determine:

a) I_{DQ} and V_{GSQ}

b) V_{GS} and V_S



1. $I_D = 0A$

$V_{GS} = -8V$

2. $V_{GS} = V_P/2 = -4V$

$I_D = 2mA$

3. $V_{GS} = 0V$, $I_D = 8mA$

$I_D = 8mA$

$V_{GS} = 0$

$R_{DS} = 29$

total load

4. $V_{GS} > 0$

$V_{GS} = 2V$

$I_D = 12.5mA$

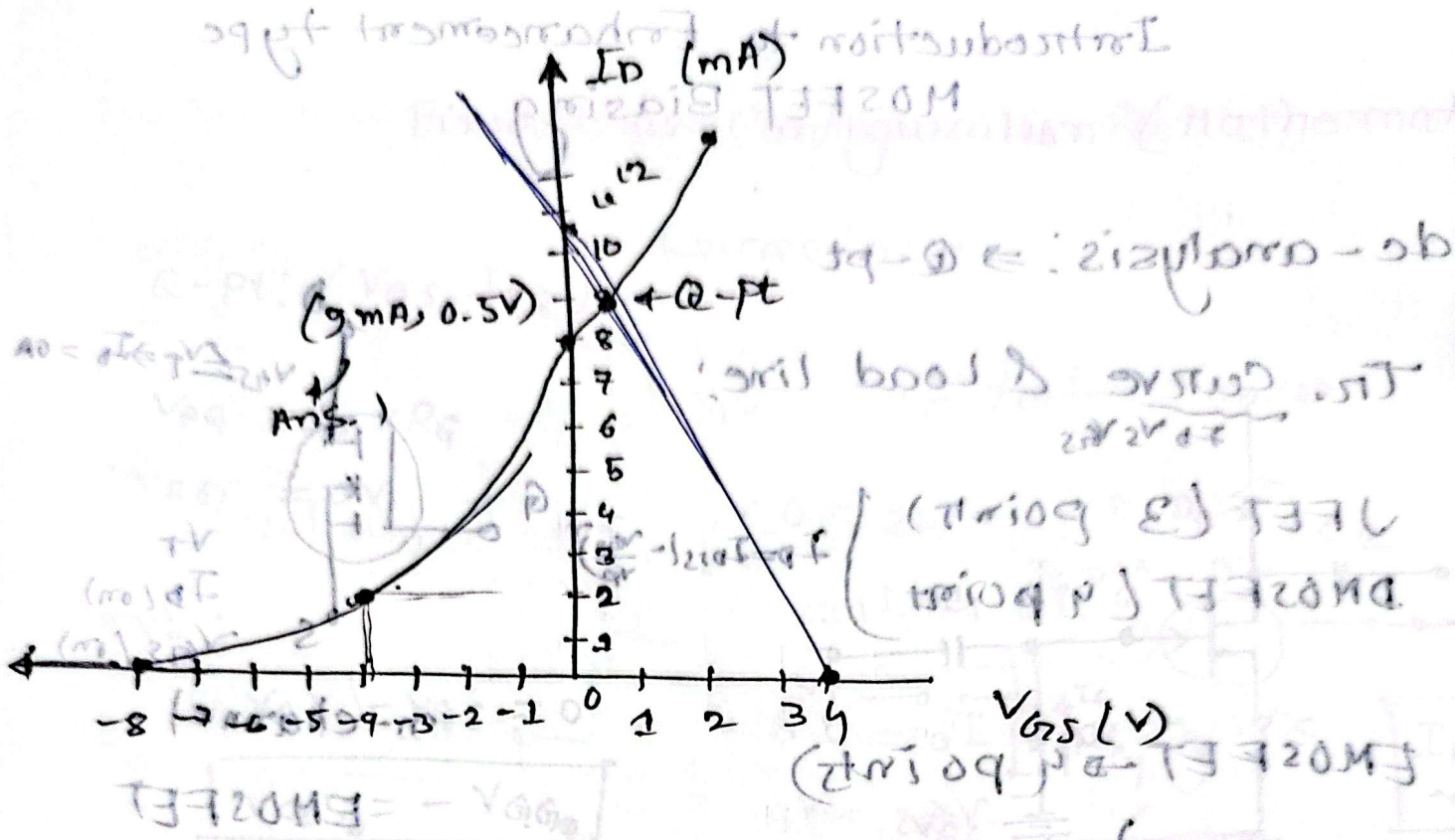
not a part

KVL: $0 - V_{GS} - I_D R_S = 0 - 9V$

$V_{GS} = 4V - (0.39k\Omega) I_D$

1. $I_D = 0A$, $V_{GS} = 4V$

2. $V_{GS} = 4V$, $I_D = 10.25 \approx 10.3mA$

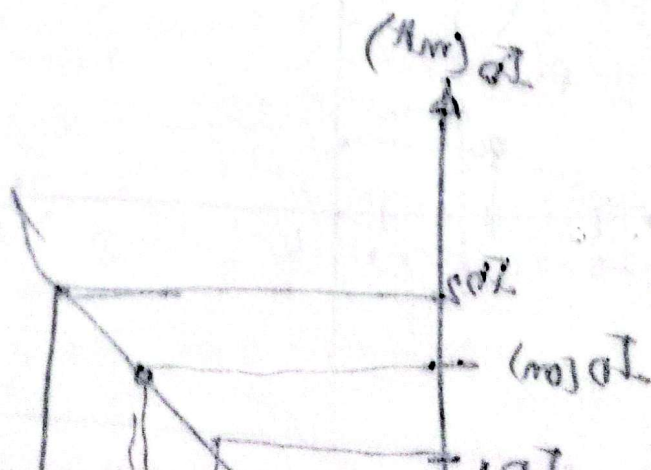


b) part: $V_{DS} \rightarrow$ KVL stating to end

$$V_{DS} = V_{DD} - I_D R_D$$

$$V_{DS} = 12 \text{ V} - I_D R_D$$

$$\frac{V_{DS}}{R_D} = \frac{12 \text{ V}}{R_D} - I_D$$



Introduction to Enhancement type MOSFET Biasing

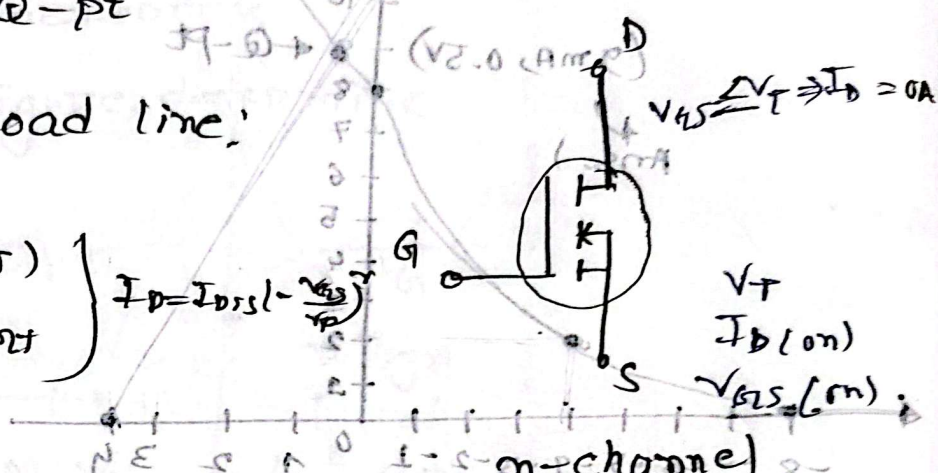
dc-analysis: $\Rightarrow$ Q-pt

Tr. Curve & Load line:

JFET (3 points)

DMOSFET (4 points)

EMOSFET (4 points)



EMOSFET

$$I_D = k(V_{GS} - V_T)^2$$

$$P_1 \equiv (V_{DS}, 0A)$$

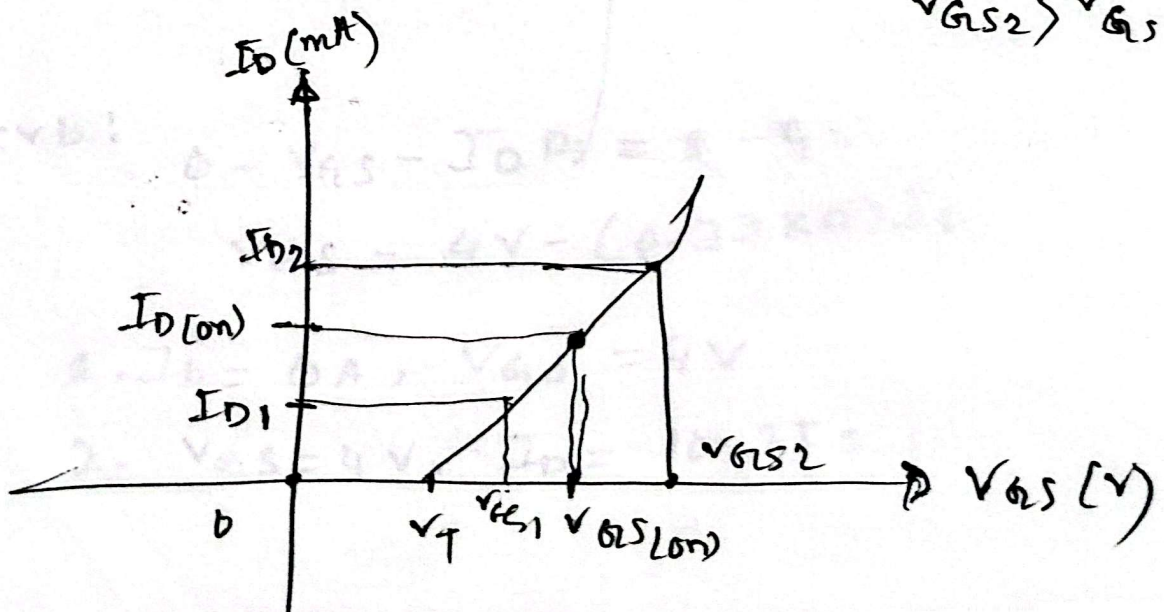
$$P_3 \equiv (V_{GS1}, I_{D1})$$

$$P_2 \equiv (V_{GS(on)}, I_{D(on)})$$

$$P_4 \equiv (V_{GS2}, I_{D2})$$

$$k = \frac{I_{D(on)}}{V_{GS(on)} - V_T}$$

$$V_T < V_{GS1} < V_{GS(on)} < V_{GS2}$$



Fixed Bias Configuration (Mathematical)

Q-pt: (V_{GSQ}, I_{DQ})

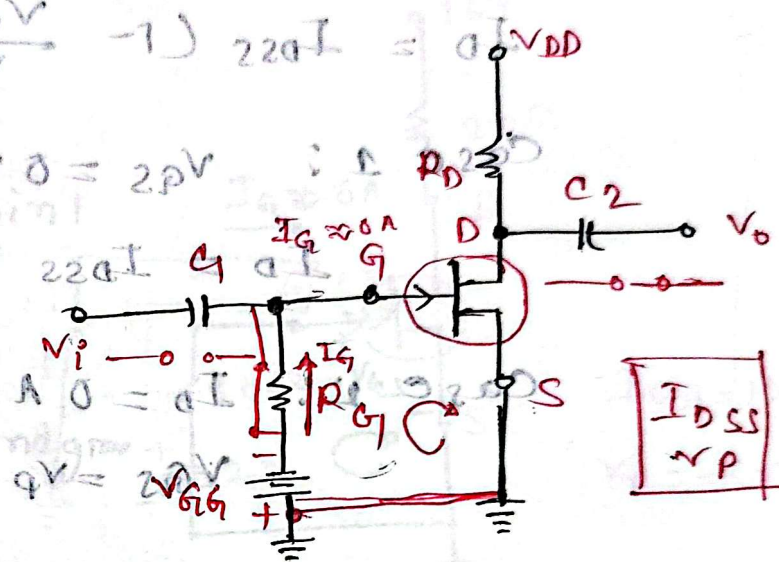
$$V_{RG} = I_G \times R_G$$

$$V_{RG} = 0V$$

KVL:

$$(-V_{GS}) - V_{GS} = 0$$

$$V_{GSQ} = -V_{GS}$$



$$I_{DQ} = I_{DSS} \left(1 - \frac{V_{GSQ}}{V_P} \right)^2$$

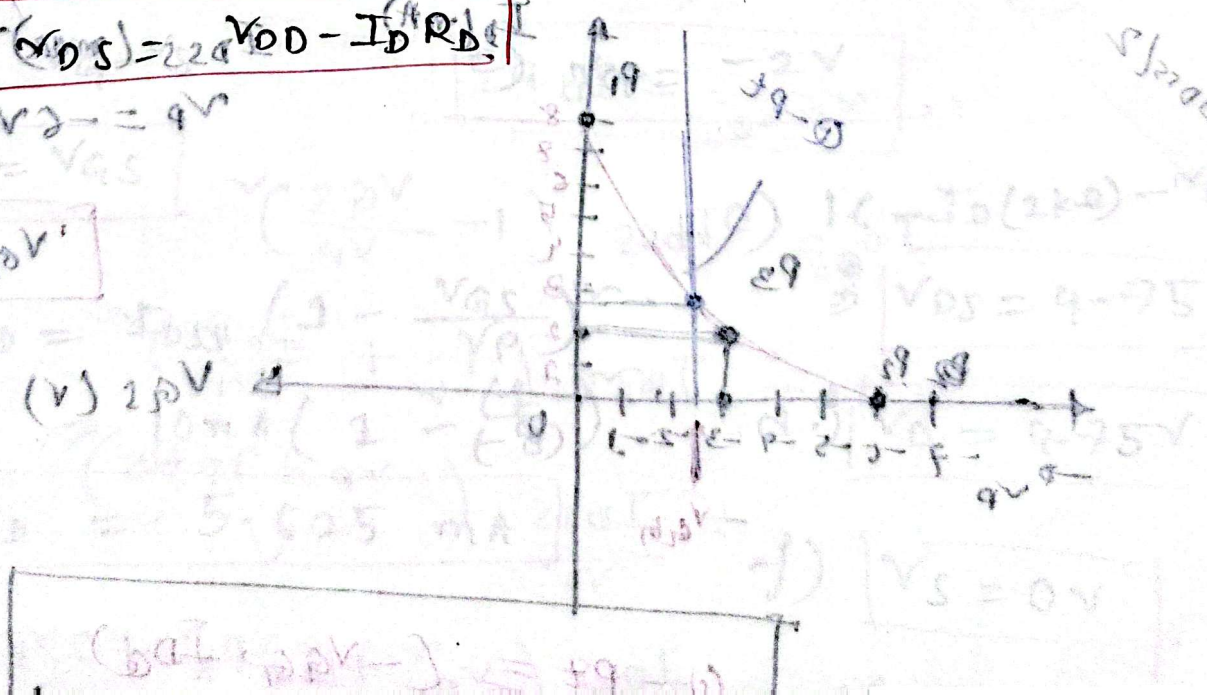
KVL: Output loop

$$V_{DD} - I_D R_D - V_{DS} = 0V$$

$$V_{DS} = V_{DD} - I_D R_D$$

$$V_{GS} = -V_{GS}$$

$$V_{GS} = -2.5V$$



Graphical Approach

n-channel

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2$$

Case 1: $V_{GS} = 0V$

$$\rightarrow P_1 (0, I_{DSS})$$

$I_D = I_{DSS}$

Case 2: $I_D = 0A$

$V_{GS} = V_p$

$$\rightarrow P_2 (V_p, 0)$$

Case 3: $V_{GS} = V_p/2$

$$I_D = I_{DSS} \left(1 - \frac{1}{2}\right)^2 = I_{DSS}/4$$

$$\rightarrow P_3 (V_p/2, I_{DSS}/4)$$

$$I_D = I_{DSS}/4$$

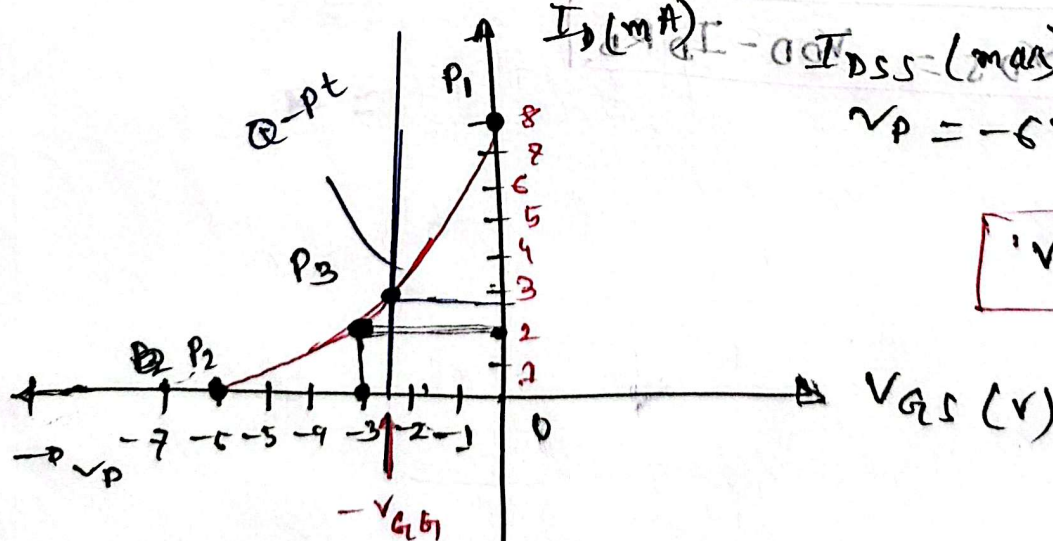
$$V_D = 20V \rightarrow 0.02A \rightarrow 20mA$$

$$I_{DSS} = 8mA$$

$$V_p = -6V$$

$$V_{GS} = -V_{GS}$$

$I_{DSS}/2$



$$Q - pt \equiv (-V_{GS}, I_{DQ})$$

Solved Problem

a. V_{GSQ}

b. I_{DQ}

c. $V_{DS} \rightarrow$ o/p volt

d. V_D

e. V_G

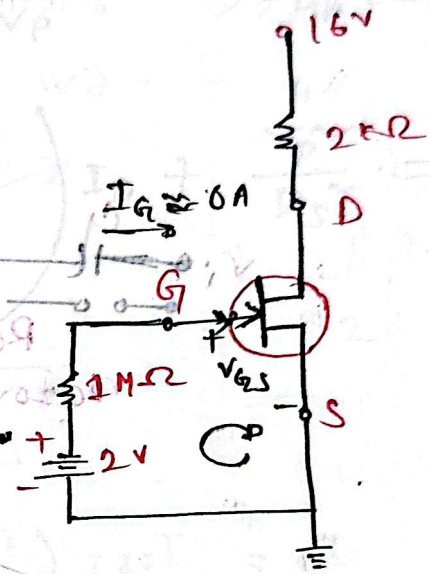
f. V_S

Potential at a point

$$V_D = \text{pot. d. between D and ground}$$

$$= V_D - 0V$$

$$= V_D$$



$I_{DSS} = 10mA$
 $V_P = -8V$

$$V_D = I_D V_{DS} - V_S$$

$$V_D = V_{DS}$$

$$V_G = V_G - 0V$$

$$= V_G - V_S$$

$$V_G = V_{GS}$$

$$-2V + (0 \times 1M\Omega) - V_{GS} = 0$$

$$V_{GS} = -2V$$

e) $V_G = -2V$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$= 10mA \left(1 - \frac{-2}{-8} \right)^2$$

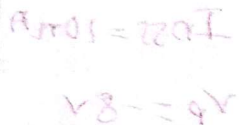
$$I_D = 5.625 mA$$

$$V_{DS} = 9.75V$$

$$V_D = 9.75V$$

$$V_S = 0V$$

one do supply



2) $I_D = I_S$ ✓

$$20V = 1V$$

$$v_0 - pV = pV$$

$$2v - pv =$$

$$2Rv = Rv$$

$$V_{DF} - p = 20V$$

$$v_0 = p \cdot f_0$$

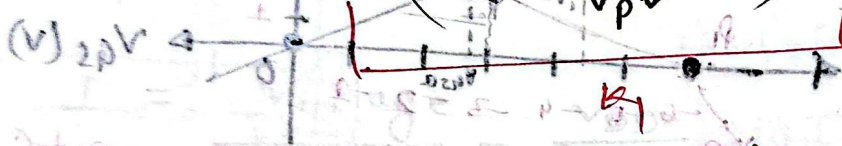
$$v_0 = 2v \quad (1)$$

$$\Rightarrow I_D = I_{DSS} + \frac{I_{DSS} R_S}{\sqrt{\rho}} I_D + \frac{2 I_{DSS} R_S}{\sqrt{\rho}} \cdot I_D$$

$$\Rightarrow 0 = I_{DSS} + \frac{I_{DSS} \cdot R_S}{V_P} I_D + \frac{2 I_{DSS} \cdot R_S}{V_P} I_D - I_D$$

$$\Rightarrow \left(\frac{I_{DSS} \cdot R_S}{V_P} \right) I_D + \left(\frac{2 I_{DSS} \cdot R_S}{V_P} - 1 \right) I_D + I_{DSS} = 0$$

$$\Rightarrow I_D + \left(\frac{2 I_{DSS} \cdot R_S - V_P}{V_P} \right) I_D + \frac{V_P}{R_S} = 0$$



$$\boxed{I_D + k_1 I_D + k_2 = 0}$$

$$V_{DD} - I_D R_D - V_{DS} = I_D R_S \Rightarrow 0 = 0$$

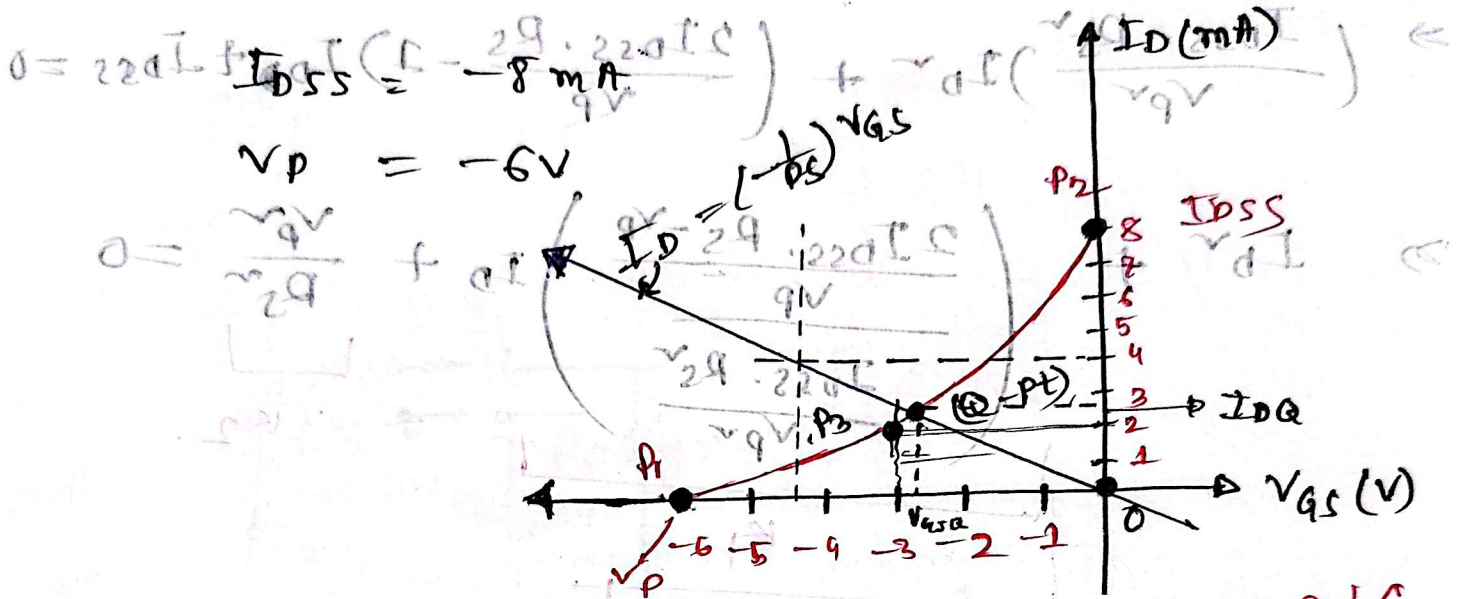
$$\boxed{V_{DS} = V_{DD} - I_D (R_D + R_S)}$$

$$V_{GS} = \frac{V_{DD}}{2} = 2.0V$$

$$V_{GS} = \frac{V_{DD}}{2} = 2.0V$$



Graphical Approach



$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$y = mx + c$
 $V_{GS} = -I_D P_S$

$P_1 \Rightarrow I_D = 0 \text{ mA}, V_{GS} = V_P$

$P_2 \Rightarrow V_{GS} = 0 \text{ V}, I_D = I_{DSS}$

$P_3 \Rightarrow V_{GS} = V_P/2 = -3 \text{ mA}$

$$I_D = \frac{I_{DSS}}{4} = 2 \text{ mA}$$

$$I_D = \left(\frac{1}{P_S} \right) V_{GS}$$

$y = mx$

$V_{GS} = 0, I_D = 0$

$$I_D = I_{DSS}/2$$

$$V_{GS} = I_{DSS} \cdot P_S$$

Voltage Divider Biasing of JFET = 20V

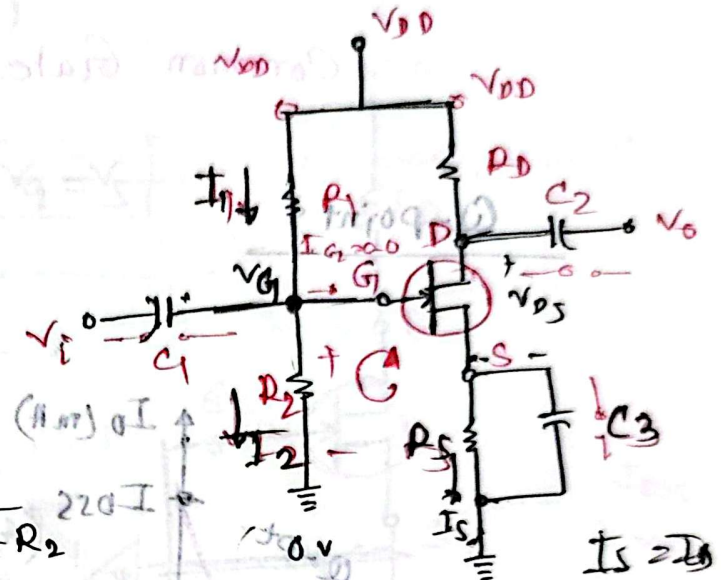
$$I_1 = I_2 + I_G$$

$$\Rightarrow |I_1 = I_2| = I$$

$$R_{eq} = R_1 + R_2$$

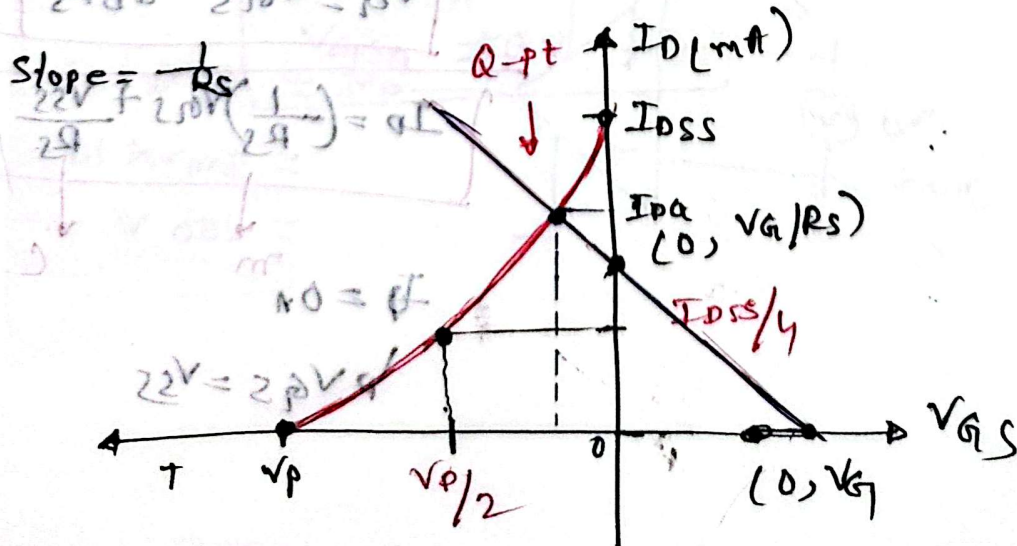
$$I = \frac{V_{DD}}{R_1 + R_2}$$

$$V_G = \frac{R_2 \times V_{DD}}{R_1 + R_2}$$



$$V_G - V_{GS} - I_D R_S = 0$$

$$I_D = \left(-\frac{1}{R_S}\right)V_{GS} + \frac{V_G}{R_S}$$

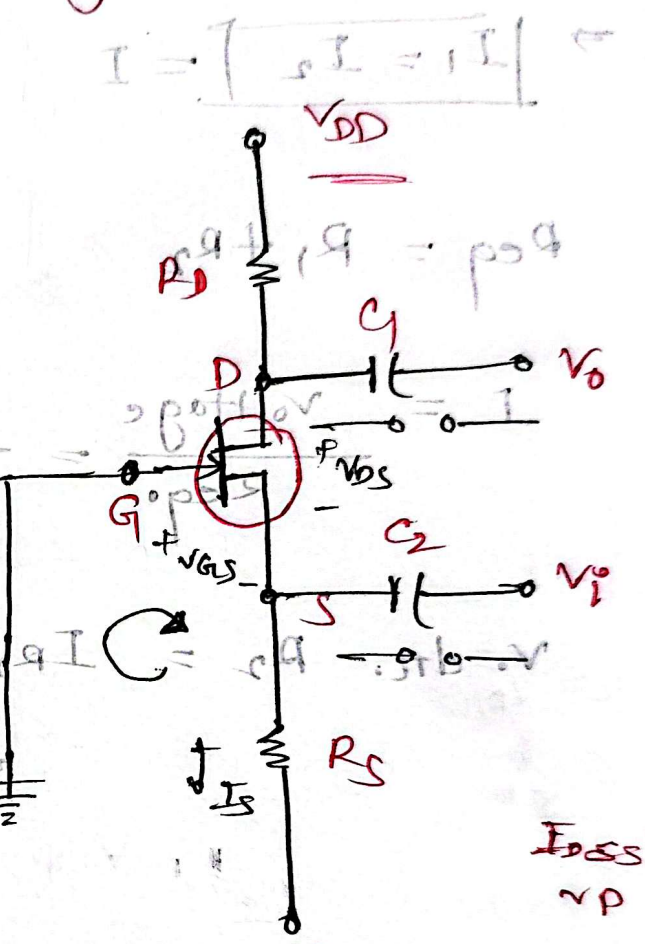
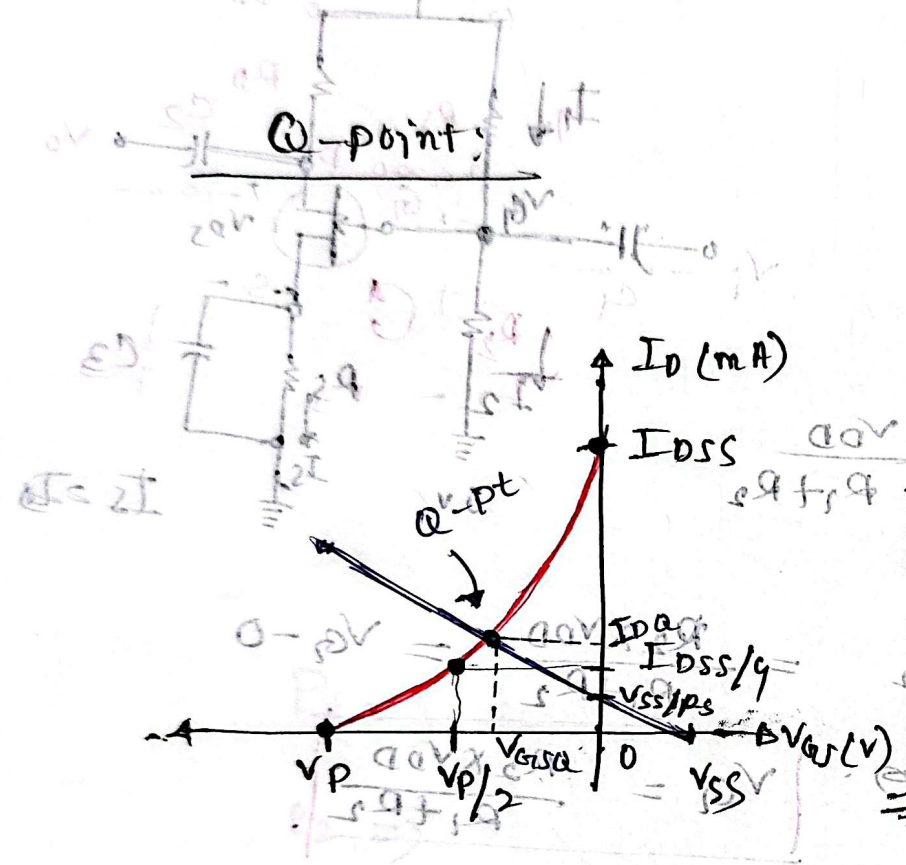


$$I_D \approx 0 A$$

$$V_{GS} = V_G$$

$$V_{DS} = V_{DD} - I_D (R_D + R_S)$$

Common Gate Configuration of JFET



$$V_{DD} - I_D R_D - V_{DS} - I_D R_S = -V_{SS}$$

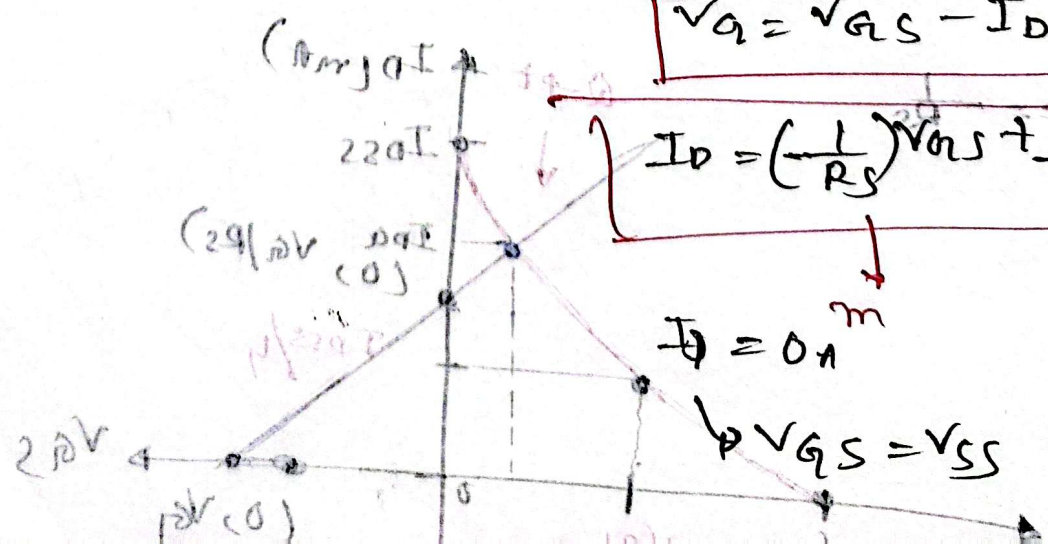
$$V_{DS} = V_{DD} + V_{SS} - I_D (R_D + R_S)$$

$$0 - V_{GS} - I_D R_S = -V_{SS}$$

$$V_G = V_{GS} - I_D R_S$$

$$I_D = \left(-\frac{1}{R_S} \right) V_{GS} + \frac{V_{SS}}{R_S}$$

$$A_0 = \frac{V_o}{V_i} = \frac{I_D R_D}{I_D R_S} = \frac{R_D}{R_S}$$

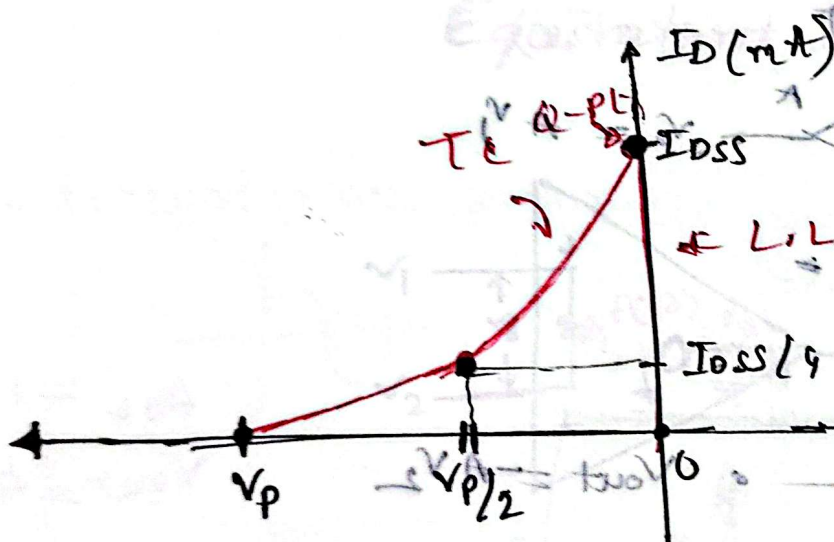
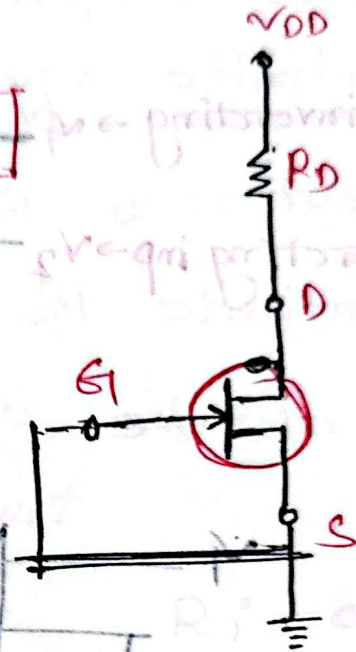
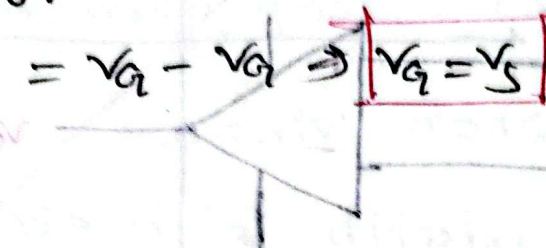


JFET Biasing (Special Case)

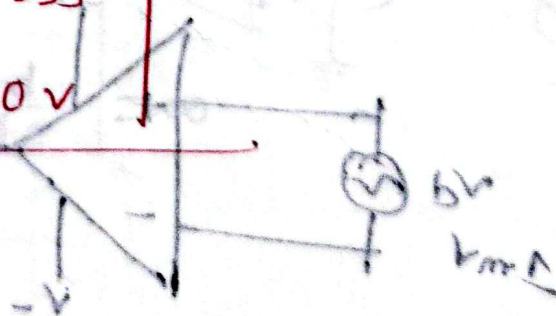
$$Q\text{-pt} \equiv (V_{GSQ}, I_{DQ})$$

$$V_{GSQ} = 0V$$

$$V_{GS} = 0 = V_G - V_S \Rightarrow V_G = V_S$$



$$\boxed{\begin{aligned} I_{DQ} &= I_{DSS} \\ V_{GSQ} &= 0V \end{aligned}}$$



$$\begin{aligned} \text{For } V_{GS} &= 0V \\ \text{For } V_{GS} &= 0V \end{aligned}$$